

Assignment 1_PERMUTATION & COMBINATION

$(x)^{25}C_{21}$

A:12650

1!	1
2!	2
3!	6
4!	24
5!	120
6!	720
7!	5040
8!	40320
9!	362880
10!	3628800

1. Find the value of

- (I) 8P_3 A: 336
 (II) 8P_8 A: 8!
 (III) ${}^{30}P_2$ A:870
 (IV) ${}^{20}P_4$ A: 116280
 (V) 7P_4 A: 840
 (VI) ${}^{12}P_5$ A: 95040
 (VII) 9P_9 A:9!
 (VIII) 9P_5 A: 15120
 (IX) ${}^{25}P_2$ A: 600
 (X) ${}^{56}P_3$ A: 166320

2 Find the value of

- (I) 7C_3 A: 35
 (II) 8C_4 A: 70
 (III) ${}^{20}C_6$ A: 38760
 (IV) 5C_1 A: 5
 (V) 6C_6 A: 1
 (VI) 7C_2 A: 21
 (VII) ${}^{10}C_4$ A: 210
 (VIII) ${}^{50}C_{50}$ A: 1
 (IX) ${}^{60}C_{58}$ A:1770

3. Find the value of $\frac{22!}{18!}$

A: 175560

4. How many numbers of 5 digits can be formed with the digits 4, 6, 7, 5, 2. (a) If no digit being repeated.

A: 120

(b) If repetition of digit be allowed, then required no. A:3125.

5. How many no. of 4 digits can be formed with the digits 5, 8, 9 & 6 no digit being repeated. A : 24

6. How many no. of 5 digits can be formed with the digits 2, 7, 0, 5, 4 ? A : 96

7. How many words can be formed by using all letters of the word, "WATCH", if repetition is not allowed ?

A : 120

8. Find the no. of words formed by using all the letter of the word HOTEL (a) so that the vowels are always together.

A : 48

(b) so that the vowels are never together. A : 72

9. How many words can be formed with the letter of the word 'TICKET' if repetition is not allowed. A : 360

10. How many words can be formed with the letter of the word "INDUSTRIAL". if repetition is not allowed.

A : 1814400

11. How many words can be formed with the letter of the word 'MEDICAL, if repetition is not allowed'

A : 5040

12. How many words can be formed with the letter of the word 'TELEPHONE', if repetition is not allowed.

A: 60480

13. Find the no. of words formed with the letter of word FIELD

(a) vowels are always together A : 48

(b) vowels are never together A : 72

(c) begins of F A: 24 (d) ends with D A: 24

(e) Has the letter E always in the middle. A: 24 (f) begins with F and ends with D. A: 6.

14. How many five letter words can be formed out of the word "RELATION" if repetition is not allowed.

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A: 6720

15. How many four letter words can be formed out of the word "SPINACH" if repetition is not allowed.

A: 840

16. How many letter words can be formed out of the word "BANKING" if repetition is not allowed.

A: 2520

17. How many letter words can be formed out of the word "COMPUTER" if repetition is not allowed.

A: 40320

18. In how many ways can 7 students be seated in a row.

Ans: 5040

19. In how many ways can 7 persons be seated around a circular table. Ans: $(7-1)! = 6! = 720$

20. In how many ways, can a (a) team of six be chosen out of nine students? (b) A team of six to be chosen out of twelve students?

Answer: (a) 84 (b) 924.

21. In how ways a committee of six members can be selected from seven men and six women

(a) Consisting of three men and three women? A:700

(b) Consisting of two men and four women? A:315

22. In a party 18 persons shake hands with every other person find the total no of handshakes in the party?

A : 153

23. In a party 35 persons shake hands with every other persons find the total no of handshakes in the party?

A : 595

24. In how many ways 7 girls and 6 boys can be seated in a row so that they are alternate? 3628800

25. In how many ways 5 girls and 4 boys can be seated in a row so that they are alternate? 2880

26. In how many ways can 3 girls and 6 boys be seated in a row so that girls are always together?

A: 30240

27. In how many ways can 4 ladies and 4 gents be seated around a circular table so that no two gents are together?

Ans: 4 Gents around a circular table = $(4-1)! = 3! = 6$ and

4 ladies in 4 ways = $4! = 24$

Total number of ways = $24 \times 6 = 144$

28. In how many ways can 5 Red beads and 7 yellow beads can be arranged so that Red beads are always together?

A: 4838400

29. In how many ways can 3 books and 8 novels can be arranged in a shelf so that novels are always together?

Answer : 967680

30. There are 7 members in a delegation which is to be sent abroad. The total no. of members is 12. In how ways can the selection be made so that a particular member is always (i) included (ii) excluded? A (i) 462 (ii) 330

31. There are 6 students in a class which is to be selected, out of 12 students. In how ways can the selection be made so that a particular student is always (i) included (ii) excluded?

(i) A :462 (ii) 462

32. Letters of the word "WATER" are arranged in such a way that all the vowels come together. Find out the total number of ways for making such arrangement. A :48

33. Letters of the word "PHRASE" are arranged in such a way that all the vowels come together. Find out the total number of ways for making such arrangement. A : 240

34. Letters of the word "NATIONAL" are arranged in such a way that all the vowels come together and repetition is not allowed. Find out the total number of ways for making such arrangement. A : 720

35. Letters of the word "PHANTOM" are arranged in such a way that all the vowels come together. Find out the total number of ways for making such arrangement. A : 1440

36. Letters of the word "PHANTOM" are arranged in such a way that vowels are never together. Find out the total number of ways for making such arrangement. A: 3600

37. Letters of the word "ORGANIZED" are arranged in such a way that all the vowels come together. Find out the total number of ways for making such arrangement. A: 17280

38. Letters of the word "MAGNITUDE" are arranged in such a way that all the vowels come together. Find out the total number of ways for making such arrangement. A : 17280

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39. Letters of the word "MAGNITUDE" are arranged in such a way that the vowels are never together. Find out the total number of ways for making such arrangement. A: 345600

40. There are five students Amit, Bina, Charu, Deepak and Ishan,

(i). In how many ways can they sit so that Bina and Charu do not sit together? A : 72

(ii). In how many ways can a committee of 3 members be formed so that Amit is always included and Ishan is always excluded? Ans 3.

41. How many numbers of five digits may be formed with the digits 3, 0, 6, 0, and 8?

Ans: 36 ways

42. How many numbers greater than 7000 can be formed with four of the digits 4, 5, 7, 8, and 9?

Ans: 72 ways

43. How many numbers greater than 500 and less than 3000 can be formed with the digits 0, 1, 2, 3, 4, 5 and 6? Ans: 300

44. A committee of some members is to be made out of a group of 9 men and 5 women. In how many different ways can it be made if the committee is to be made according to the following stipulation?

(a) the committee of 4 must include at least one women.

A : 875

(b) the committee of 8 must have exactly four men and four women . A : 630

45: A committee of 6 students is to be formed out of 4 boys and 7 girls. In how many ways can it be done so that the no. of girls is not less than the no. of boys?

Ans: 441

46. From 6 Professors and 4 Lecturers a committee of 5 is to be formed. In how many ways can this be done if the committee is to include at least one Lecturer?

Ans: 246

47. There are 12 hats out of which 7 are red and 5 are blue. In how many ways a person can choose 6 hats such that at least 4 of them are red? A: 462

48. A man has to select 3 cakes from 3 chocolate cakes and 2 vanilla cakes. In how many different ways he can make the selection according to the following stipulation

A) When he selects exactly 2 chocolate cakes and 1 vanilla cake. Ans: 6

B) When he selects at least 1 vanilla cake.

Ans: 9

49. In how many ways 2 girls and 4 boys can be seated in a row so that (a) girls are always together? A: $5! \times 2! = 240$

(b) boys are always together? A: $3! \times 4! = 144$

50. In how many ways 3 girls and 5 boys can be seated in a row so that girls are always together and boys are also always together? A: $2! \times 3! \times 5! = 1440$.